



UNIVERSITY OF NAIROBI

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DEPARTMENT OF MECHANICAL AND MANUFACTURING ENGINEERING

Design Specification Sheet: Lightweight FDM Lattice Structures

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1. Functional Requirements

- **Load Bearing:** The lattice specimen must support quasi-static compressive loads without buckling globally prior to unit cell yield.
- **Periodicity:** The specimen must feature a regular repeating unit cell (e.g., Simple Cubic or Body-Centered Cubic) capable of being mathematically scaled.
- **Manufacturability:** The specimen must be printable utilizing standard Fused Deposition Modeling (FDM) technology without requiring internal breakaway supports that could damage struts during post-processing.
- **Envelope Limits:** The core lattice volume must fit entirely within a 60 mm × 60 mm × 60 mm bounding box.
- **Boundary Conditions:** The specimen must include solid top and bottom caps to ensure a uniform distribution of the compressive load from the testing machine platens.
- **Parametric Integrity:** The CAD design must be fully parametric and editable to accommodate rapid topological iterations.

2. Performance Targets

- **Target Specimen Dimensions:** 60 mm $\times$ 60 mm $\times$ 60 mm core lattice, flanked by two 2 mm thick solid caps, resulting in a total height of 64 mm.
- **Mass Reduction:** The final lattice must achieve a mass reduction of at least 50% when compared to a solid PLA block of equivalent outer volume.
- **Simulation Convergence:** The finite element model must demonstrate less than a 5% variance in peak displacement between the final two successive mesh refinement steps.
- **Print Viability:** Zero major layer delaminations, missing struts, or mid-print failures across the finalized specimens.
- **Experimental Repeatability:** The force-displacement profiles of three identically printed specimens must yield a standard deviation of less than 10% in peak compressive strength, displaying consistent physical failure modes.

3. Material Baselines & Sensitivity Analysis Framework

Because FDM printing inherently introduces micro-voids, variable layer adhesion, and surface roughness, utilizing idealized bulk material properties in FEA guarantees discrepancies. To properly calibrate the digital model against physical reality, the following sensitivity analysis framework will be executed computationally:

- **Baseline Literature Properties (PLA):**
 - Young’s Modulus (E): 2.55 GPa
 - Poisson’s Ratio (ν): 0.35
 - Density (ρ): 1.24 g/cm³
- **Modulus Sensitivity Calibration:** To account for FDM porosity, the FEA static simulation will be run using a degraded matrix of effective Young’s Moduli: $E = 2.0$ GPa, $E = 2.25$ GPa, and $E = 2.5$ GPa.
- **Strut Geometry Offset:** Surface roughness and under-extrusion effectively reduce the load-bearing cross-section of printed struts. Simulation variants will be tested with effective strut diameters offset by -0.1 mm and -0.2 mm from the nominal CAD dimensions to model the FDM manufacturing penalty.